

- 2 The International Space Station (ISS) orbits the Earth at a height of 4.1×10^5 m above the Earth's surface. The radius of the Earth is 6.37×10^6 m.

- (a) Both the ISS and the astronauts inside it are in free fall. Explain why this makes the astronauts feel weightless

There is no contact force between the astronaut and the floor of the space station.

.....[1]

- (b) (i) Calculate the value of the gravitational field strength g at the height of the ISS above the Earth.

$$g_0 = \frac{GM}{R^2}, \quad g_0 R^2 = GM \quad [1]$$

$$g = \frac{GM}{(R+h)^2}, \quad g = \frac{g_0 R^2}{(R+h)^2} = \frac{9.81 \times (6.37 \times 10^6)^2}{(6.37 \times 10^6 + 4.1 \times 10^5)^2} = 8.66 \text{ N kg}^{-1} \quad [1]$$

$$g = \dots\dots\dots \text{N kg}^{-1} \quad [2]$$

- (ii) State the value of the centripetal acceleration of ISS at this height.

$$a_c = \dots\dots\dots 8.66 \dots\dots\dots \text{m s}^{-2} \quad [1]$$

- (iii) The speed of the ISS in its orbit is 7.7 km s^{-1} . Show that the period of the ISS in its orbit is 92 minutes.

$$vT = 2\pi(R+h) \quad [1]$$

$$7700T = 2\pi(6.37 \times 10^6 + 4.1 \times 10^5)$$

$$T = 92.2 = 92 \text{ min} \quad [1]$$

[2]

- (iv) The ISS is in a low Earth orbit. Suggest an advantage of this orbit as compared to higher orbits.

It requires less fuel to launch, and hence it is less expensive.

It is easier access for maintenance and repair.

Higher resolution of images captured by ISS of features on the surface of the Earth.

[1]

- (c) The ISS has arrays of solar cells on its wings. These solar cells charge batteries which power the ISS. The wings always face the Sun.

7% of the energy of the sunlight incident on the cells is stored in the batteries. The total area of the cells facing the solar radiation is 2500 m^2 . The intensity of solar radiation at the orbit of the ISS is 1.4 kW m^{-2} outside of the Earth's shadow and zero inside it. The ISS passes through the Earth's shadow for 35 minutes during each orbit.

By reference to **(b)(iii)**, calculate the average power delivered to the batteries during one orbit.

Power stored in batteries

$$= 0.07 \times \text{Intensity} \times \text{Area} = 0.07 \times 1.4 \times 10^3 \times 2500 = 2.45 \times 10^5 \text{ W} \quad [1]$$

$$\text{Cells are in the Sun for } 92 - 35 = 57 \text{ min} \quad [1]$$

$$\text{Average power} = \frac{57}{92} \times 2.45 \times 10^5 = 1.5 \times 10^5 \text{ W} \quad [1]$$

average power =W [3]

[Total: 10]